Lecture Notes on

The Fermi Paradox

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These notes summarize the third Astrophysics Club meeting of the 2026-27 year. The question for the meeting: if intelligent life is possible, why haven't we seen it? We build up the numbers for how fast a civilization could fill the galaxy, show why the silence is a paradox, and then look at the ways out: the Drake equation's uncertainty, the Great Filter, and what it would mean if we are alone.

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1 Background Facts

1.1 Time

- The Milky Way is about 13 billion years old. Earth formed about 4.6 billion years ago, so the galaxy had roughly 9 billion years of head start on us.
- Our technology is about 100 years old. A civilization with a head start of just 1 million years would be 10,000 times more advanced in age than we are.

1.2 Count

- The galaxy has about 10^{11} stars, and surveys suggest a large share of them host a rocky planet in the habitable zone. That gives 10^{10} to 10^{11} candidate worlds.
- If only 1 in 1,000,000 of those worlds ever produces a technological civilization, that is 10^4 to 10^5 civilizations.

1.3 Distance

- The galaxy is about 100,000 light-years across, so light needs 100,000 years to cross it.
- A ship at $0.1c$ (10% of light speed) needs 1,000,000 years, which is only about 0.01% of the galaxy's age.

2 Exponential Colony Growth

What if each colony launches more colonies? After n generations, the number of colonies is

$$N = k^n,$$

which over time becomes

$$N(t) = k^{t/\tau}.$$

- k is the number of new colonies each colony produces.
- n is the number of generations.
- τ is the time between generations.

If $k = 2$, the count doubles every generation:

Generation	Colonies
10	1,024
20	1,048,576
30	1,073,741,824

Nobody has to make the whole trip across the galaxy in one hop. Each colony is a new launch point. If τ is 10,000 years, 30 generations is only 300,000 years, which already beats the single-ship crossing time by a lot. **Growth rate beats ship speed, but this can't last.**

3 Not Infinite Stars

Growth has to stop when there are no more stars to colonize. The logistic equation captures this:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right).$$

- N is the number of colonized systems.
- r is the expansion rate.
- K is the maximum number of accessible systems, around 10^{11} stars.

The brake. The $(1 - N/K)$ term slows growth to zero as N approaches K. On the same axes, the exponential curve shoots off the top of the chart, while the logistic curve flattens out at K.

4 Expansion as a Wave

Add spreading through space to the logistic growth and you get the Fisher equation:

$$\frac{\partial N}{\partial t} = D\nabla^2 N + rN\left(1 - \frac{N}{K}\right).$$

- **Spread plus growth.** D measures how far colonies spread per unit time. The second term is the same logistic growth from the last section.
- **The Fisher equation** is the same math used for spreading populations and invasive species.
- **A constant-speed front.** It predicts a colonization front that moves at a constant speed,

$$v = 2\sqrt{rD}.$$

- **Crossing time.** The galaxy crossing time is the diameter divided by v. A front moving at $0.01c$ to $0.001c$ crosses 100,000 light-years in 10^7 to 10^8 years.

5 Why It's a Paradox

The Milky Way is about 13×10^9 years old.

- **Optimistic.** With a crossing time of 10^6 years, that is $13 \times 10^9 \nabla \cdot 10^6 \approx 13{,}000$ expansion times.
- **Pessimistic.** With the wave from the last section, 10^8 years, it is still $13 \times 10^9 \nabla \cdot 10^8 \approx 130$ expansion times.
- **Light lag** can explain a delay of thousands of years, not billions. A civilization 5,000 light-years away is a 10,000 year round trip, which is nothing on this timescale.

And yet there is no confirmed evidence of anyone.

Definition (The Fermi Paradox)**.** If technological civilizations should have had enough time to become widespread, why don't we see evidence of them?

6 The Drake Equation

$$N = R_* \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

- N: number of technologically advanced civilizations in the Milky Way.
- R_*: rate of formation of stars in the galaxy.
- f_p: fraction of those stars with planetary systems.
- n_e: number of planets per solar system with an environment suitable for life.
- f_l: fraction of suitable planets on which life actually appears.
- f_i: fraction of life-bearing planets on which intelligent life emerges.
- f_c: fraction of those planets that develop interstellar communication.
- L: length of time such civilizations release detectable signals into space.

Some of these factors are known to within a factor of 10. Others are not known to within a factor of 10^{30}.

7 Uncertainty of the Constants

Take f_l, the chance that life starts on a habitable planet. Nobody knows it, but a plausible range is 10^{-30} to 1. Treat every power of ten in that range as equally likely.

- The **mean** is about 0.014. With the mean, 10^{11} planets $\times 0.014$ gives about 10^9 worlds with life.
- The **median** is 10^{-15}. With the median, $10^{11} \times 10^{-15}$ gives 10^{-4} worlds.
- Once f_l drops below 10^{-11}, we expect fewer than one living world in the whole galaxy. That covers 19 of the 30 powers of ten, so there is about a 63% chance we are in that zone.

On a log-scale number line from 10^{-30} to 1, the median (10^{-15}) and the mean (about 10^{-2}) sit far apart, and everything below 10^{-11} means fewer than one living world.

8 Maybe We Are Alone

In 2018, Anders Sandberg, Eric Drexler, and Toby Ord of Oxford's Future of Humanity Institute argued that the paradox comes from Drake-style equations treating highly uncertain numbers as if they were known.

- **How they did it.** Instead of one best guess per term, they used the full range of uncertainty from the literature for every term, including models of how life could start for f_l.
- **Result.** After updating on the fact that we see no one, they found a 53% to 99.6% chance that we are alone in the galaxy, and a 39% to 85% chance that we are alone in the observable universe.

Silence isn't surprising. It is a likely outcome. The paradox came from multiplying best guesses.

9 The Great Filter

Economist Robin Hanson theorized that at least one step on the path from dead matter to a civilization that spreads across the galaxy must be extremely unlikely.

Suppose becoming a spacefaring civilization requires several steps:

$$A \to B \to C \to D \to E.$$

The total chance is the product of the chances of each step:

$$P = p_1 \times p_2 \times p_3 \times p_4 \times p_5.$$

- If each step has a 10% chance, $P = (0.1)^5 = 10^{-5} = 0.001\%$.
- With 10^{10} candidate worlds, that still gives about 10^5 civilizations.
- To get down to one or fewer, the product has to be around 10^{-10}, which is five steps at 1% each.

Five steps that each look passable can multiply into something wildly unlikely.

10 Where Is the Filter?

Picture life as a staircase. Chemistry becomes self-replicating life, then complex cells, then multicellular life, then big brains and tools, then a species that starts to leave its planet. We are on that last step. The next steps are our solar system, other stars, and the whole galaxy. We see no one on the upper steps, so something must block the way. That block is the filter.

10.1 Behind Us?

The hard step could be:

- **Life starting at all.** There is no consensus on how likely that is.
- **The complex cell.** As far as we know, it happened exactly once, when one cell took in another and the two stayed together as partners.
- **Intelligence.** A big brain is expensive and does not guarantee survival.

10.2 Ahead of Us?

- **Only a roadblock.** A disaster we can recover from, even if recovery takes a million years.
- **A true filter** has to be so devastating that it has ended almost every civilization that reached it.
- **Candidates:** nuclear war, engineered pathogens, runaway AI, nanotechnology, or damaging our own planet.

This is why finding life elsewhere would be bad news. The more common and advanced life is, the more likely the filter is ahead of us.

11 So Where Is Everybody?

We started with a simple question: if the galaxy is this old and this big, where is everybody? The math said the galaxy has had time to fill up many times over, so silence needs an explanation.

- **Life is rare.** Maybe intelligent life is extraordinarily rare, and we are one of the few hard steps' winners.
- **Short-lived.** Maybe technological civilizations are short-lived and burn out before they spread.
- **Hard to expand.** Maybe interstellar expansion is much harder than our toy model, or civilizations have no reason to expand.
- **Wrong evidence.** Maybe we are looking for the wrong evidence, or we are early, with a deadline on the way.

Or maybe we have just been multiplying guesses. Once the uncertainty is included, an empty galaxy is a likely outcome.

If we are alone, then everything that lives on this planet may be all the life there is, and it is up to us to keep it going.